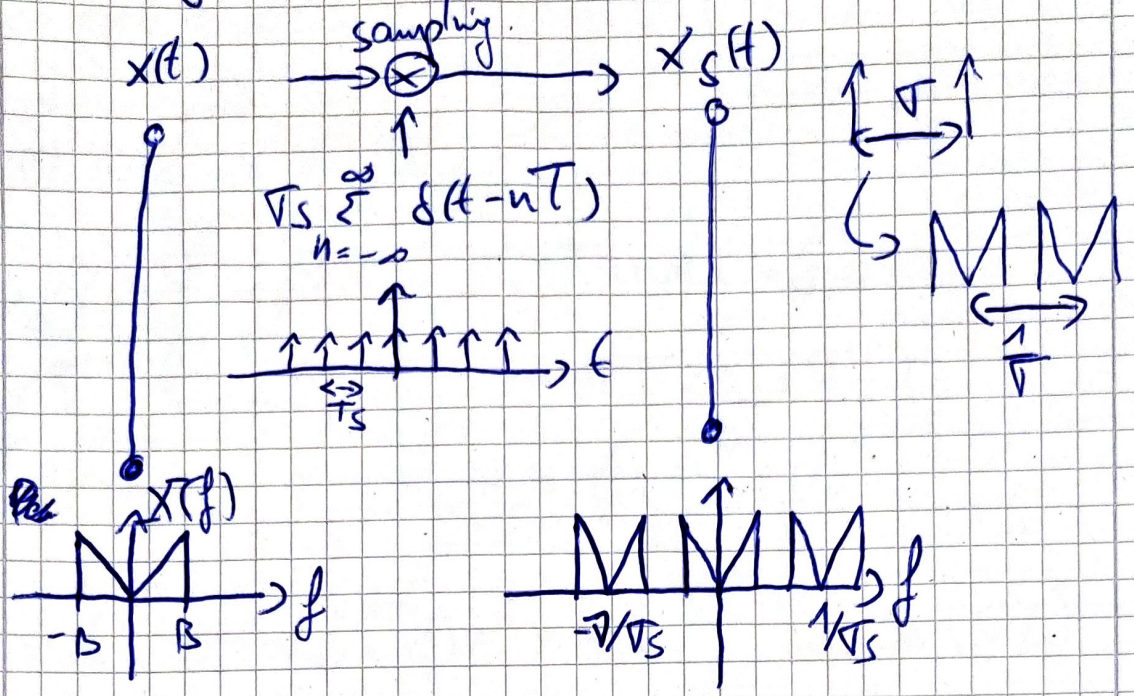
Signale & Systeme

Sampling



Perfect reconstruction of the analog signal is possible if the sampling frequency $f_s = 1/T_s$ meets:

$$f_s \geq 2B$$

The analog signal can be perfectly reconstructed using a low pass filter

Signale & Systeme

Sampling

$$x(t) = \frac{2}{T} si\left(\frac{2\pi t}{T}\right) + \frac{1}{T} si^2\left(\frac{\pi t}{T}\right)$$

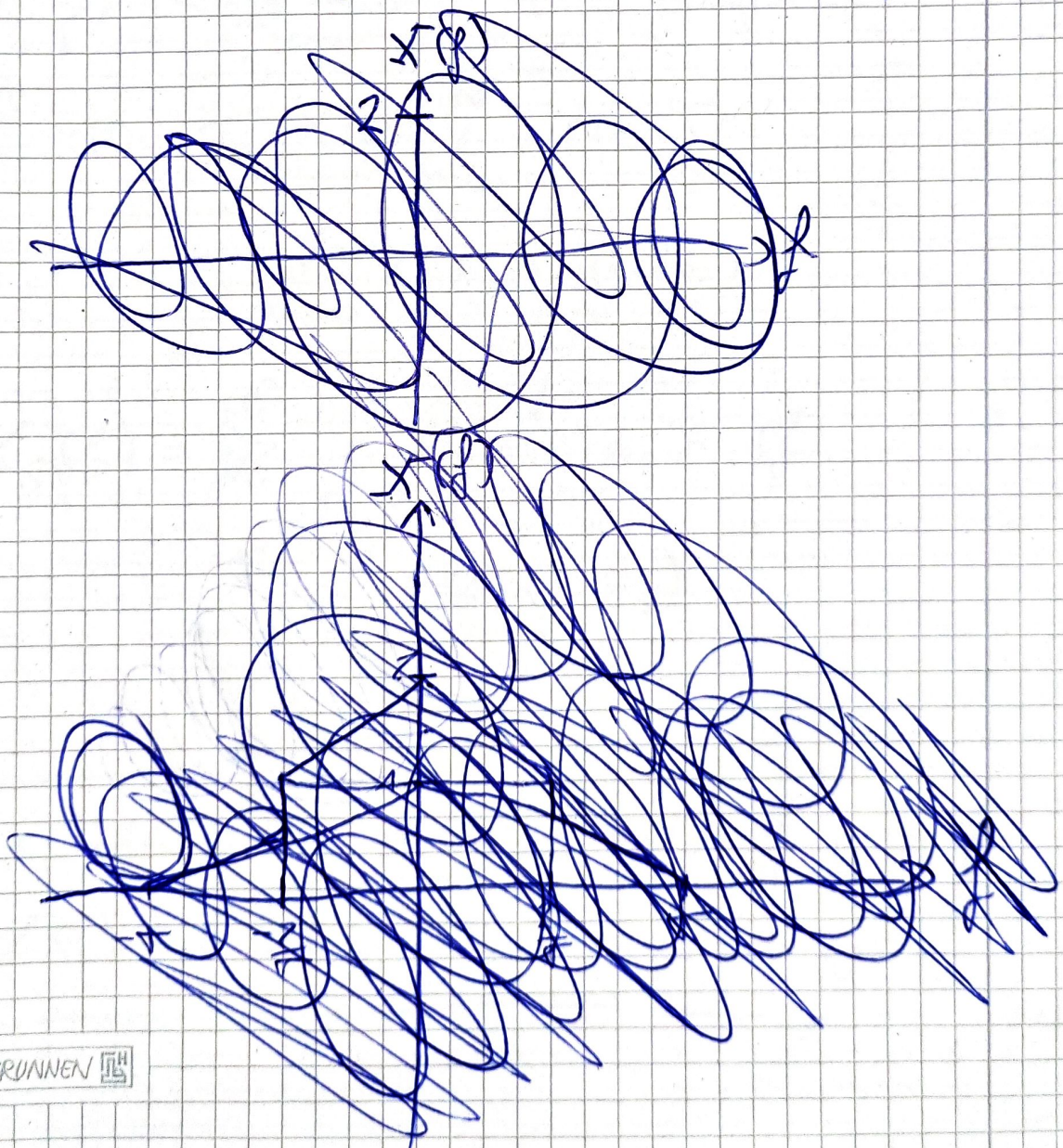
a) Calculate $X(f)$

$si\left(\frac{\pi t}{T}\right) \circ\!\!-\!\!\bullet\; T \cdot rect(Tf)$

$si^2\left(\frac{\pi t}{T}\right) \circ\!\!-\!\!\bullet\; |T| \wedge (Tf)$

$x(at) \circ\!\!-\!\!\bullet\; \frac{1}{|a|} X(f/a)$

$$x(t) \circ\!\!-\!\!\bullet\; X(f) = rect\left(\frac{Tf}{2}\right) + \wedge(Tf)$$

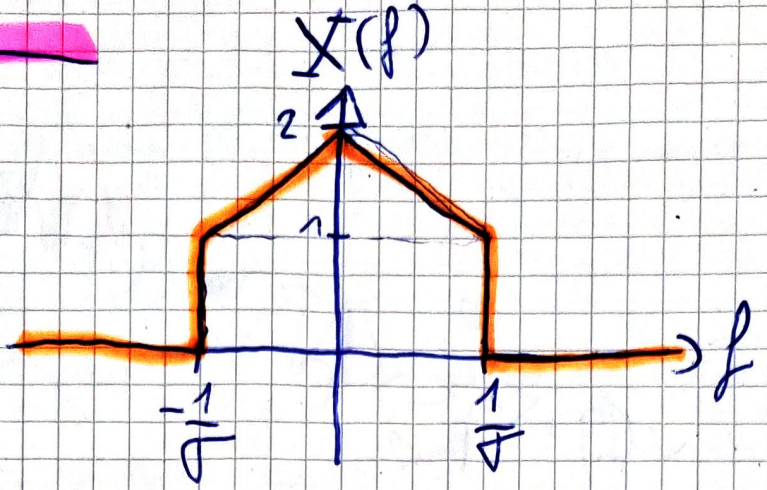


BRUNNEN

Signale & Systeme

Sampling

a)



b) $X(k) = T_S \cdot X(kT_S)$, $f_S = \frac{1}{T_S}$

$X_S(f) = X(f) \cdot \frac{1}{T_S} \sum_{n=-\infty}^{\infty} \delta(f - \frac{k}{T_S})$

$X(k) = T_S \cdot X(t) \cdot \sum_{k=-\infty}^{\infty} \delta(n - kT_S)$

$= T_S \sum_{k=-\infty}^{\infty} X(kT_S)\, \delta(n - kT_S)$

$X_S(f) = \frac{f_S}{T_S} \sum_{n=-\infty}^{\infty} \delta(f - \frac{k}{T_S}) * X(f)$

$= \sum_{n=-\infty}^{\infty} X(f - \frac{k}{T_S})$

$X(t)\,\delta(t-t_0) = X(t_0)\,\delta(t-t_0)$

$X(t) * \delta(t-t_0) = X(t-t_0)$

BRUNNEN

Signale & Systeme

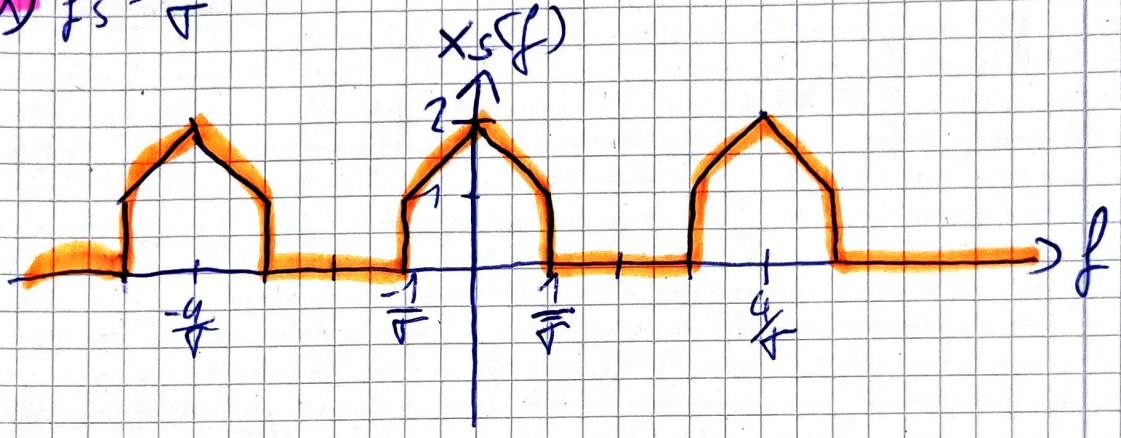
Sampling

c) Which minimum sampling frequency $f_{s,min}$ is required?

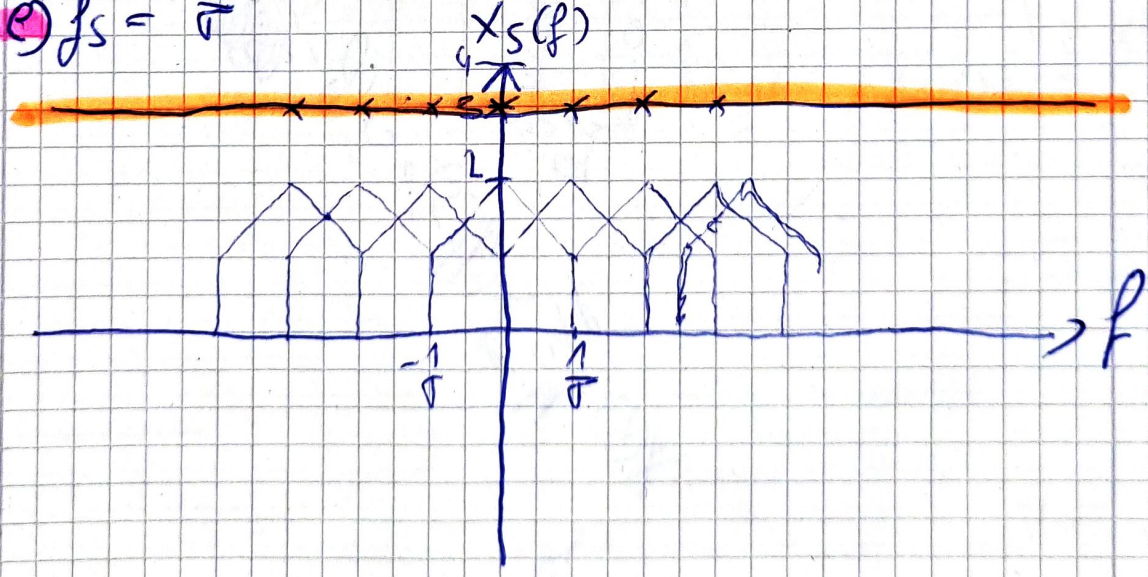
$$B = \frac{1}{T}$$

$$f_s \geq 2B = \frac{2}{T}$$

d) $f_s = \frac{4}{T}$



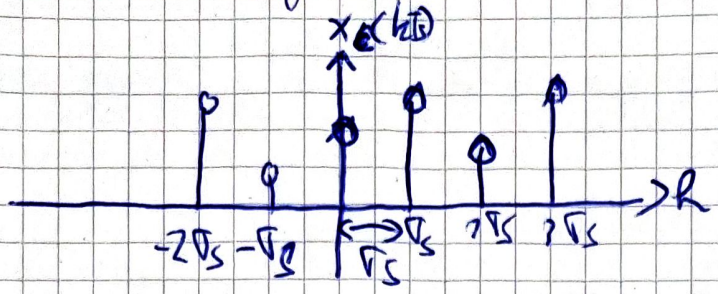
e) $f_s = \frac{1}{T}$



BRUNNEN

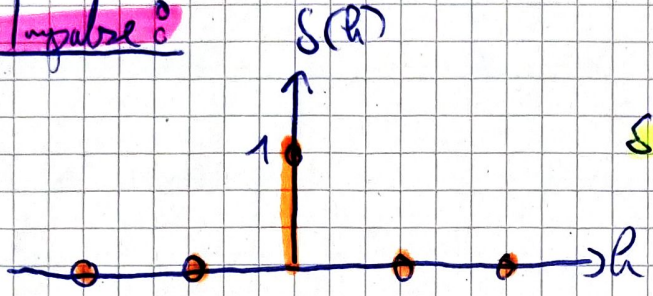
Signale & Systeme

Discrete-Time Signals



$f_s = \frac{1}{T_s}$, sampling frequency , $-\infty < k < \infty$

Dirac Impulse:



$$\delta(k) = \begin{cases} 1, & k = 0 \\ 0, & k \neq 0 \end{cases}$$

Step Function:

$$\varepsilon(k) = \begin{cases} 1, & k \geq 0 \\ 0, & k < 0 \end{cases}$$

BRUNNEN TH

Signale & Systeme

Discrete Time Fourier Transform (DTFT)

→ non periodic & discrete time

$$X(\Omega) = \sum_{k=-\infty}^{\infty} x[k]\, e^{-j\Omega k} \quad ; \; \Omega = 2\pi f / f_s$$

The discrete-time Fourier transforms a discrete-time signal $x(k)$ into the frequency domain

Inverse DTFT: $x(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\Omega)\, e^{j\Omega k}\, d\Omega$

the DTFT spectrum $X(\Omega)$ of a discrete time signal is periodic with period 2π.

Rules of DTFT

$x(k-k_0) \quad \circ\!\!-\!\!\bullet \quad e^{-j\Omega k_0} X(\Omega)$

$e^{j\Omega_0 k} x(k) \quad \circ\!\!-\!\!\bullet \quad X(\Omega - \Omega_0)$

$x(k) \cdot y(k) \quad \circ\!\!-\!\!\bullet \quad \frac{1}{2\pi} X(\Omega) * Y(\Omega)$

$x(k) * y(k) \quad \circ\!\!-\!\!\bullet \quad X(\Omega) \cdot Y(\Omega)$

$x^*(k) \quad \circ\!\!-\!\!\bullet \quad X^*(-\Omega)$

$x^*(-k) \quad \circ\!\!-\!\!\bullet \quad X^*(\Omega)$

$x(k) = x_{Re}(k) + x_{Ro}(k) + j x_{Ie}(k) + j x_{Io}(k)$

$X(\Omega) = X_{Re}(\Omega) + X_{Ro}(\Omega) + j X_{Ie}(\Omega) + j X_{Io}(\Omega)$

BRUNNEN

Signale & Systeme

DTFT

a) Verify $x(k-k_0) \circ\!\!-\!\!\bullet\; e^{-j\Omega k_0} X(\Omega)$

~~$X(\Omega) = \sum_{k=-\infty}^{\infty} x(k-k_0)$~~

$\Leftrightarrow$

$$x(k-k_0) = x(k) * \delta(k-k_0) \circ\!\!-\!\!\bullet\; X(\Omega) \cdot Y(\Omega)$$

$$Y(\Omega) = \sum_{k=-\infty}^{\infty} \delta(k-k_0) e^{-j\Omega k} = \sum_{k=-\infty}^{\infty} \delta(k-k_0) e^{-j\Omega(k_0)}$$

$$= e^{-j\Omega k_0}$$

b) Verify $x^*(k) \circ\!\!-\!\!\bullet\; X^*(-\Omega)$

$$X(\Omega) = \sum_{k=-\infty}^{\infty} x(k) e^{-j\Omega k} = \sum_{k=-\infty}^{\infty} x(k) (\cos(-\Omega k) + j\sin(-\Omega k))$$

$(\dots)^*$

$$= \sum_{k=-\infty}^{\infty} x^*(k) (\cos(\Omega k) - j\sin(\Omega k) = X^*(-\Omega)$$

Signale & Systeme

DTFT

e) $x(k) = a^k \varepsilon(k), \quad |a| < 1$

$$X(\Omega) = \sum_{k=-\infty}^{\infty} a^k \varepsilon(k) e^{-j\Omega k}$$

$$= \sum_{k=0}^{\infty} a^k e^{-j\Omega k} = \sum_{k=0}^{\infty} \left(\frac{a}{e^{j\Omega}}\right)^k$$

Geometrische Reihe $|q| < 1$: $\sum_{k=0}^{\infty} a_0 q^k = \frac{a_0}{1-q}, \quad |q| < 1$

$$= \frac{1}{1 - \frac{a}{e^{j\Omega}}} = \frac{e^{j\Omega}}{e^{j\Omega} - a}$$

d) Calculate the IDTFT of $X(k)$

$$X(\Omega) = 1 + 2\cos^2(\Omega/2)$$

$$x(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\Omega) e^{j\Omega k} \, d\Omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left(1 + 2\cos^2\left(\frac{\Omega}{2}\right)\right) e^{j\Omega k} \, d\Omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j\Omega k} + e^{j\Omega k} + \cos(\Omega) e^{j\Omega k} \, d\Omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} 2e^{j\Omega k} + \cos(\Omega) e^{j\Omega k} \, d\Omega$$

Signale & Systeme

DTFT:

d) $X(\Omega) = 1 + 2\cos^2\left(\frac{\Omega}{2}\right)$

$= 1 + 1 + \cos(\Omega)$

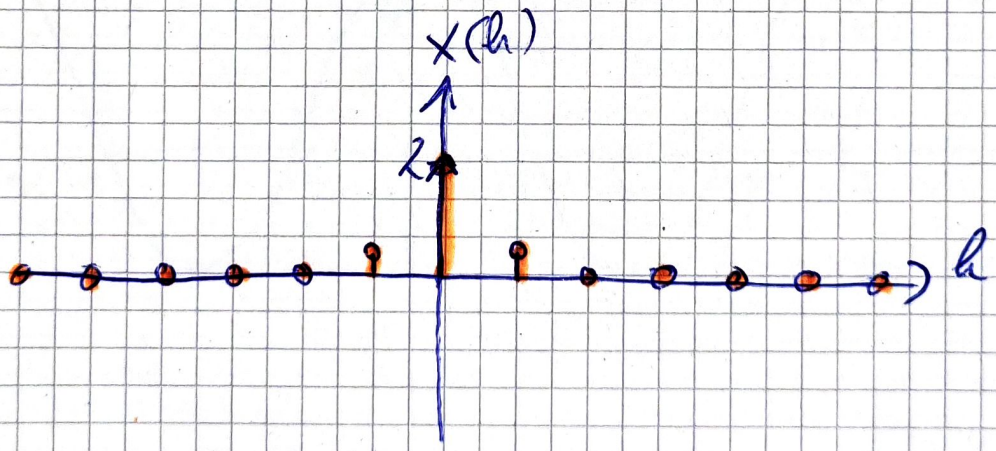
$= 2 + \cos(\Omega)$

$= 2e^{j\Omega \cdot 0} + \frac{1}{2}\left(e^{j\Omega} + e^{-j\Omega}\right)$

$= 2e^{j\Omega \cdot 0} + \frac{1}{2}e^{j\Omega \cdot 1} + \frac{1}{2}e^{j(-1)\Omega}$

$$X(\Omega) = \sum_{k=-\infty}^{\infty} x(k)\, e^{-j\Omega k}$$

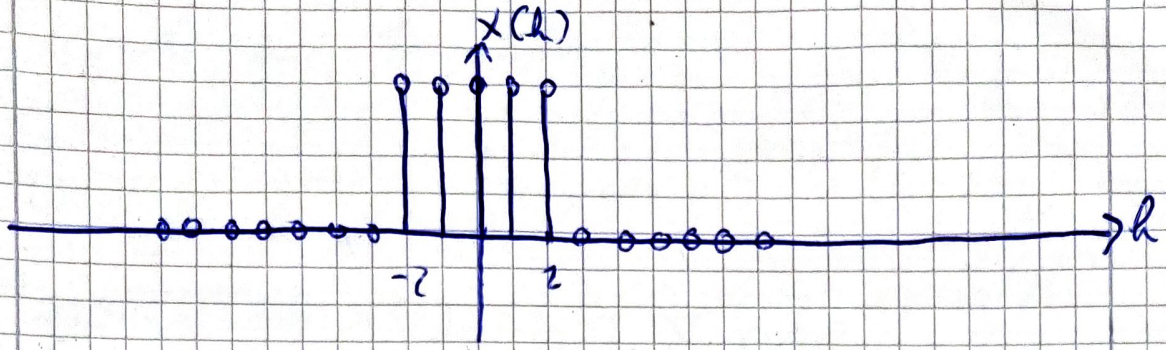
$$x(k) = \begin{cases} 2, & k = 0 \\ 1/2, & k = \pm 1 \\ 0, & \text{else} \end{cases}$$



$$x(k) = 2\,\delta(k) + \frac{1}{2}\left(\delta(k-1) + \delta(k+1)\right)$$

Signale & Systeme

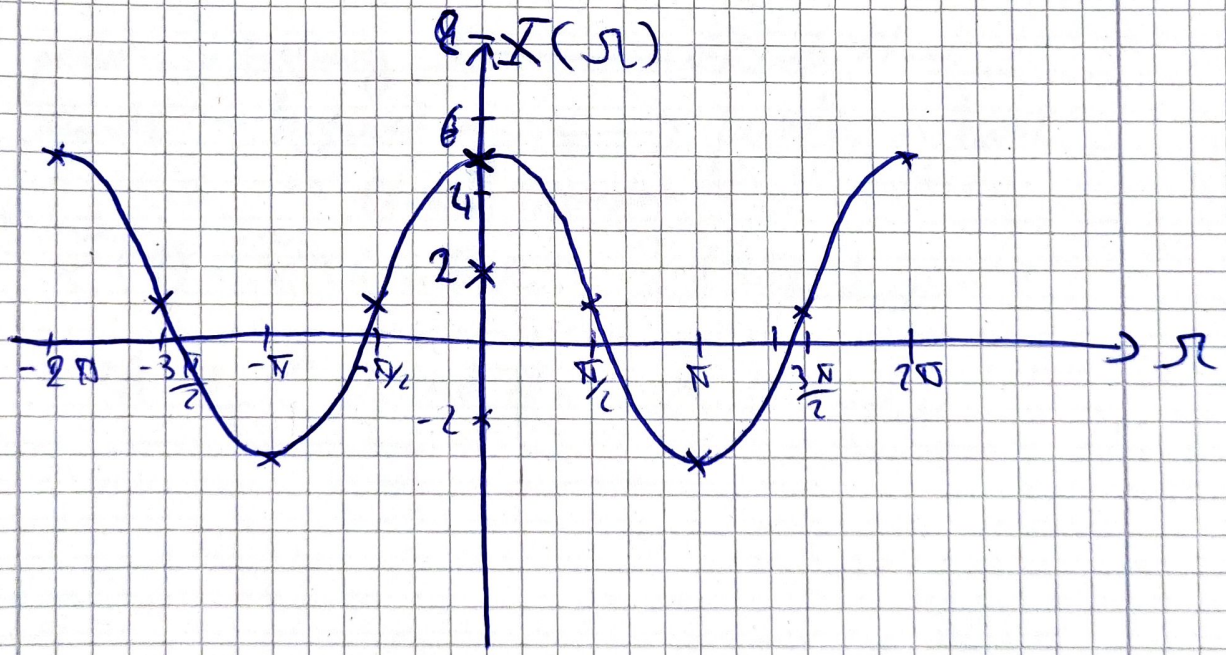
DTFT & DFT



a) Calculate the DTFT $X(\Omega)$

$$X(\Omega) = \sum_{k=-\infty}^{\infty} x(k) e^{-j\Omega k} = e^{-j\Omega 2} + e^{-j\Omega 1} + 1 + e^{j\Omega 1} + e^{j\Omega 2}$$

$$= 1 + 2\cos(\Omega) + 2\cos(2\Omega)$$



Signale & Systeme

Discrete-Fourier Transform (DFT)

discrete-time & periodic

DFT: $\underline{X}(n) = \sum_{k=0}^{N-1} x(k)\, e^{-j 2\pi n k / N}$

Inverse DFT: $x(k) = \frac{1}{N} \sum_{n=0}^{N-1} \underline{X}(n)\, e^{j 2\pi n k / N}$

Discrete in time and frequency domain!
Finite number of samples N.

Cyclic Properties of DFT

$\underline{X}(n) = \underline{X}(n + i N)$; $x(k) = x(k + i N)$

periodic in frequency ⟵ discrete time
discrete in frequency ⟶ periodic in time

Periodic Extension

$\tilde{x}(k) = \tilde{x}(k \bmod N) = \tilde{x}(k)_{\bmod N}$

$\underline{\tilde{X}}(n) = \underline{\tilde{X}}(n \bmod N) = \underline{\tilde{X}}(n)_{\bmod N}$

BRUNNEN

Signale & Systeme

DFT-Matrix

$$\begin{bmatrix} X(0) \\ \vdots \\ X(N-1) \end{bmatrix} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & e^{-j2\pi \frac{1\cdot 1}{N}} & \cdots & e^{-j2\pi \frac{1\cdot(N-1)}{N}} \\ \vdots & \vdots & & \vdots \\ 1 & e^{-j2\pi \frac{(N-1)\cdot 1}{N}} & \cdots & e^{-j2\pi \frac{(N-1)^2}{N}} \end{bmatrix} \begin{bmatrix} x(0) \\ \vdots \\ x(N-1) \end{bmatrix}$$

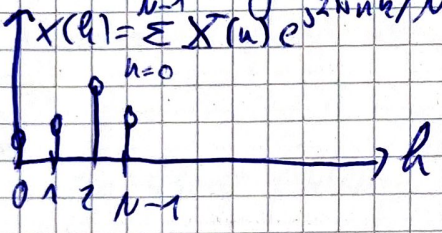
$\underline{\underline{X}}$ = F · x

DFT: $\underline{\underline{X}} = F x$

IDFT: $x = \frac{1}{N} F^H \underline{\underline{X}}$

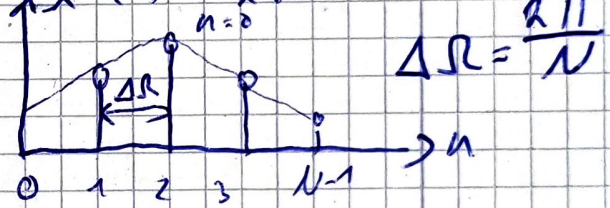
Zero Padding

$$x(k) = \sum_{n=0}^{N-1} X(n) e^{j2\pi nk/N}$$



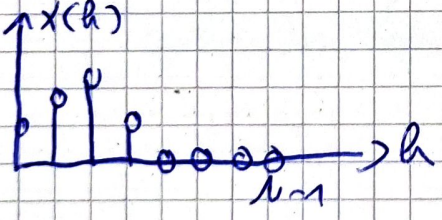
$$X(n) = \sum_{n=0}^{N-1} x(k) e^{-j2\pi nk/N}$$

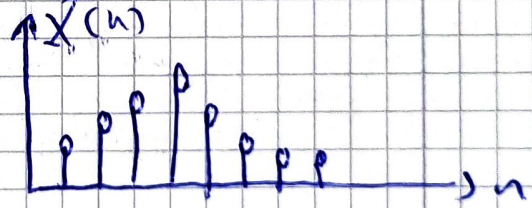
$$\Delta\Omega = \frac{2\pi}{N}$$



!

$N \rightarrow 2N$



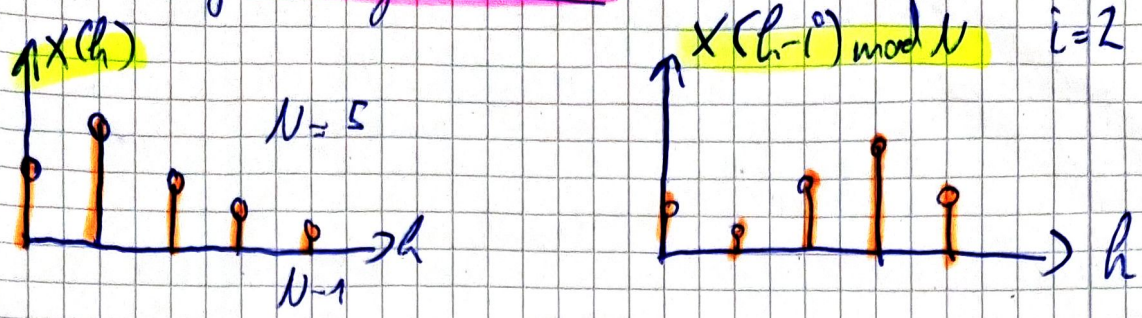


Zero padding in time ⟶ Interpolation in frequency

Interpolation in time ⟵ Zero padding in frequency

Signale & Systeme

DFT - Cyclic shift



A cyclic shift can also be regarded as the result of shifting the periodically repeated signal $\tilde{x}(n)$.

$$x(n-i) \bmod N \quad \circ\!\!-\!\!\bullet \quad e^{-j2\pi i n/N} X(u)$$

Non Cyclic Vs Cyclic Discrete Convolution

In general different results ↯

But

1.) Zero padding : $N \geq N_1 + N_2 - 1$

↳ Same results within $0 \leq k < N$ as for non-cyclic convolution

Signale & Systeme

DFT - Properties

$$X(n) = \sum_{k=0}^{N-1} x(k)\, e^{-j 2\pi k n / N}$$

$$x(k) = \frac{1}{N} \sum_{k=0}^{N-1} X(n)\, e^{j 2\pi k n / N}$$

$\alpha x_1(k) + \beta x_2(k) \quad \overset{DFT}{\circ\!\!-\!\!\bullet} \quad \alpha X_1(n) + \beta X_2(n)$

$x(k-i) \bmod N \quad \circ\!\!-\!\!\bullet \quad e^{-j 2\pi n i / N} X(n)$

$e^{j 2\pi i k / N} x(k) \quad \circ\!\!-\!\!\bullet \quad X(n-i) \bmod N$

Cyclic convolution

$$\sum_{q=0}^{N-1} x_1(k-q)_{\bmod N}\, x_2(q) \quad \circ\!\!-\!\!\bullet \quad X_1(n) X_2(n)$$

$x_1(q) x_2(q) \quad \circ\!\!-\!\!\bullet \quad \frac{1}{N} \sum_{q=0}^{N-1} X_1(n-q)_{\bmod N}\, X_2(q)$

$$x(k) = x_{Re}(k) + x_{Ro}(k) + j x_{Ie}(k) + j x_{Io}(k)$$

$$X(n) = X_{Re}(n) + X_{Ro}(n) + j X_{Ie}(n) + j X_{Io}(n)$$

Even signal: $x_e(k) = x_e(-k)_{\bmod N} = x_e(N-k)$

Odd signal: $x_o(k) = -x_o(-k)_{\bmod N} = -x_o(N-k)$

$$x^*(k) \quad \circ\!\!-\!\!\bullet \quad X^*(N-k)$$

$$x^*(N-k) \quad \circ\!\!-\!\!\bullet \quad X^*(n)$$

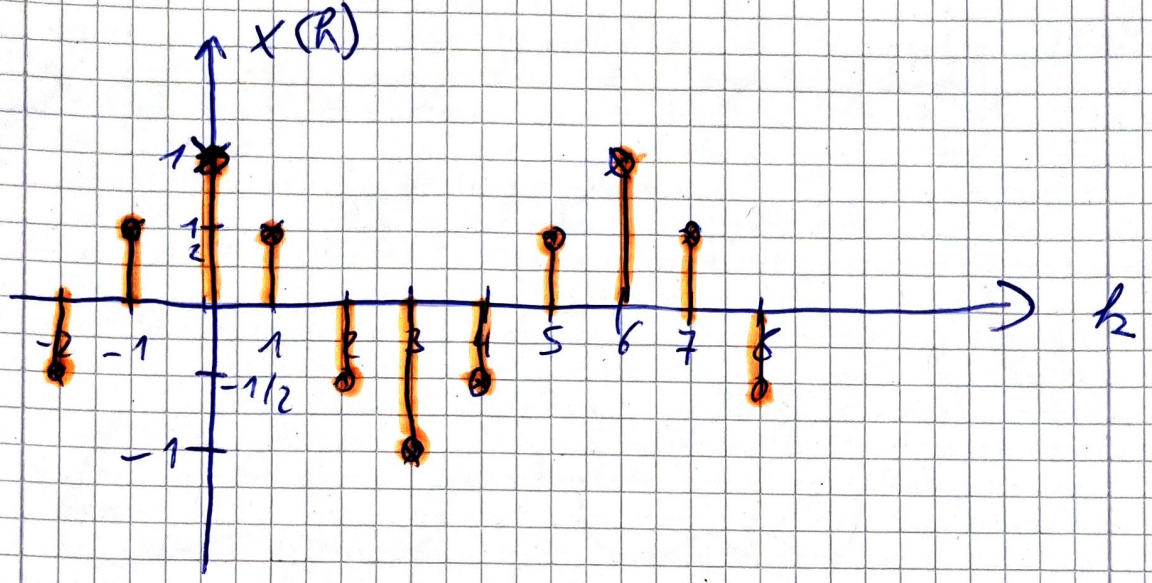
$$\sum_{k=0}^{N-1} e^{-j\frac{2\pi}{N} n k} = \begin{cases} N, & n = l \cdot N,\ l \in \mathbb{Z} \\ 0, & \text{else} \end{cases}$$

Signale & Systeme

Discrete Fourier Transform

$$X(k) = \cos\left(\frac{2\pi k}{6}\right)$$

a) Draw $x(k)$ $\quad -2 \le k \le 8$



b) $N = 6$

$$X(n) = \sum_{k=0}^{5} x(k)\, e^{-j2\pi nk/N}$$

$$= 1 + \frac{1}{2}e^{-j2\pi n/6} - \frac{1}{2}e^{-j4\pi n/6} - e^{-j6\pi n/6} - \frac{1}{2}e^{-j8\pi n/6} + \frac{1}{2}e^{-j10\pi n/6}$$

$$= 1 + \frac{1}{2}e^{-j\frac{\pi n}{3}} - \frac{1}{2}e^{-j\frac{2\pi n}{3}} - e^{-j\pi n} - \frac{1}{2}e^{-j\frac{4\pi n}{3}} + \frac{1}{2}e^{-j\frac{5\pi n}{3}}$$

$$= 1 + \frac{1}{2}\left(\cos\left(\frac{\pi}{3}n\right) - j\sin\left(\frac{\pi}{3}n\right)\right) - \cos(\pi n)$$

Signale & Systeme

DFT:

$$\sum_{k=0}^{N-1} e^{-j\frac{2\pi}{N}nk} = \begin{cases} N & , n = l \cdot N, \; l \in \mathbb{Z} \\ 0 & , \text{else} \end{cases}$$

b) $N = 6$

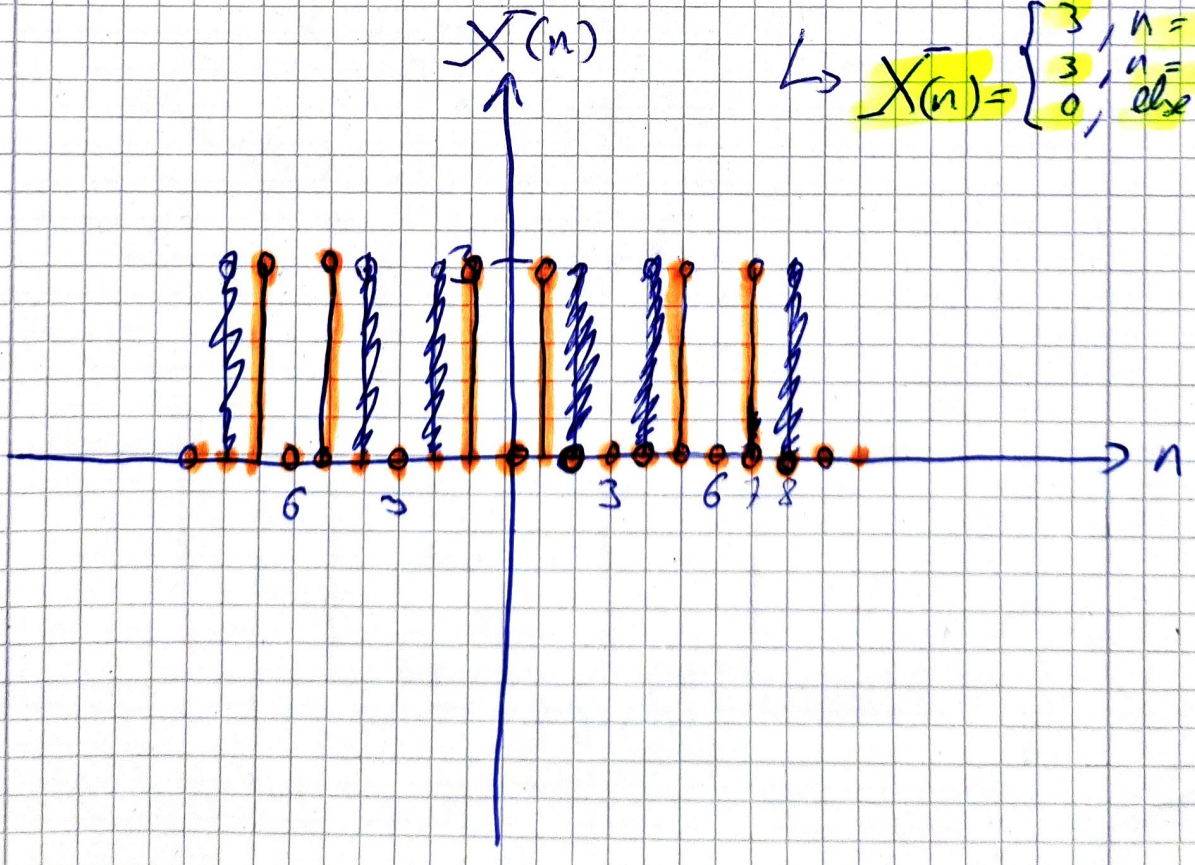
$$\underline{X}(n) = \sum_{k=0}^{5} x(k) e^{-j2\pi nk/N} = \sum_{k=0}^{5} \cos\left(\frac{2\pi k}{6}\right) e^{-j2\pi nk/N}$$

$$= \sum_{k=0}^{5} \frac{1}{2}\left(e^{j\frac{2\pi k}{6}} + e^{-j\frac{2\pi k}{6}}\right) e^{-j2\pi nk/6}$$

$$= \sum_{k=0}^{5} \frac{1}{2} e^{-j\frac{2\pi k(n-1)}{6}} + \frac{1}{2} e^{-j\frac{2\pi(n+1)}{6}}$$

$$* = \begin{cases} \frac{6}{2} & , n-1 = l \cdot N^{6} \\ 0 & , \text{else} \end{cases} = \begin{cases} \frac{6}{2} & , n+1 = l \cdot N^{6} \\ 0 & , \text{else} \end{cases}$$

$$\hookrightarrow \underline{X}(n) = \begin{cases} 3 & , n = 6l-1 \\ 3 & , n = 6l+1 \\ 0 & , \text{else} \end{cases}$$



Signale & Systeme

DFT:

a) $N = 12$

$$X(n) = \sum_{k=0}^{11} \cos\left(\frac{2\pi k}{6}\right) e^{-j\frac{2\pi nk}{12}}$$

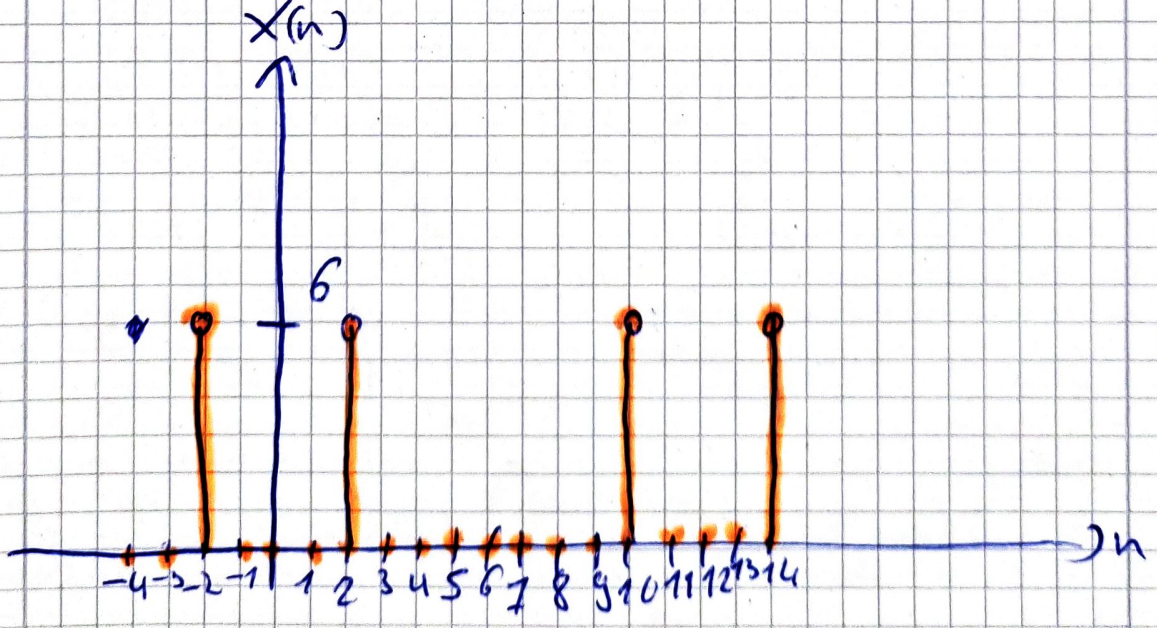
$$= \sum_{k=0}^{11} \frac{1}{2}\left(e^{j\frac{2\pi k}{6}} + e^{-j\frac{2\pi k}{6}}\right) e^{-j\frac{2\pi nk}{12}}$$

$$= \sum_{k=0}^{11} \frac{1}{2} e^{-j\frac{2\pi k(n-2)}{12}} + \frac{1}{2} e^{-j\frac{2\pi k(n+2)}{12}}$$

$$= \begin{cases} 6 \,;\, n-2 = 12l \\ 0, \text{else} \end{cases} \qquad = \begin{cases} 6 \,;\, n+2 = 12l \\ 0 \,;\, \text{else} \end{cases}$$

$n = 12l + 2 \quad \overset{!}{=} \quad 12l - 2$

$$X(n) = \begin{cases} 6, & n = 12l + 2 \\ 6, & n = 12l - 2 \\ 0, & \text{else} \end{cases}$$



Signale & Systeme

DFT:

a) $N = 16$

$$X(n) = \sum_{k=0}^{15} \frac{1}{2}\left(e^{-j\frac{2\pi k}{6}} + e^{-j\frac{2\pi k}{6}}\right) e^{-j\frac{2\pi nk}{16}}$$

$$= \sum_{k=0}^{15} \frac{1}{2} e^{-j\frac{2\pi k(3n-8)}{48}} + \frac{1}{2} e^{-j\frac{2\pi k(3n+8)}{48}}$$

$3n \pm 8 = 48 \cdot l \rightarrow$ no integer solution

Signale & Systeme

Multicarrier Transmission II

$$x(t) = x_1(t) + x_2(t) = d_1 \cos(2\pi f_1 t) + d_2 \cos(2\pi f_2 t) \; ; \; f_2 = 2 f_1$$

discrete time!

$$x(k) = x(t = k T_S) \quad ; \; T_S = \frac{1}{f_S}$$

a) Which range of values for f_S is allowed, such that, in principle, $x(t)$ can be generated from its sampled values $x(k)$ error free?

Highest frequency of $x(t)$: $B = \frac{2\pi f_2}{2\pi} = 2 f_1$

↳ $f_S \geq 4 f_1$

we set $f_S = 8 f_1$

b) $\tilde{x}(k) = x(t) \cdot \sum_{n=-\infty}^{\infty} \delta(k - n \cdot T_S)$

$$= \sum_{n=-\infty}^{\infty} x(k - n T_S)$$

$$= \sum_{n=-\infty}^{\infty} d_1 \cos(2\pi f_1 (k - n T_S)) + d_1 \cos(2\pi f_2 (k - n T_S))$$

BRUNNEN

Signale & Systeme

Multicarrier Transmission II

b) $x(k) = x(t = k T_s)$

$$= d_1 \cos\left(\frac{2\pi f_1 k}{8 f_1}\right) + d_2 \cos\left(\frac{2\pi \cdot 2 \cdot f_1 k}{8 \cdot f_1}\right)$$

$$= d_1 \cos\left(\frac{\pi k}{4}\right) + d_2 \cos\left(\frac{\pi k}{2}\right)$$

$$= d_1 \left(e^{j\frac{\pi k}{4}} + e^{-j\frac{\pi k}{4}}\right) + d_2 \left(e^{j\frac{\pi k}{2}} + e^{-j\frac{\pi k}{2}}\right)$$

c) inverse N-DFT ($N = 8$), derive a formula for the real output signal $\hat{x}(k)$

$$e^{j\frac{2\pi n k}{N}} = e^{j\frac{2\pi (n-N) k}{N}}$$

$$\hat{x}(k) = \sum_{n=0}^{7} X(n) e^{j\frac{2\pi n k}{8}}$$

$$= 0 + \frac{d_1}{2} e^{j\frac{2\pi \cdot 1 \cdot k}{8}} + \frac{d_2}{2} e^{j\frac{2\pi \cdot 2 k}{8}} + \frac{d_2}{2} e^{j\frac{2\pi \cdot 6 k}{8}} + \frac{d_1}{2} e^{j 2\pi \frac{7 \cdot k}{8}}$$

$$= \frac{d_1}{2} e^{j\frac{2\pi k}{8}} + \frac{d_1}{2} e^{j\frac{2\pi k (7-8)}{8}} + \frac{d_2}{2} e^{j 2\pi \frac{2 \cdot k}{8}} + \frac{d_2}{2} e^{j\frac{2\pi (6-8) k}{8}}$$

$$= \frac{d_1}{2}\left(e^{j\frac{\pi k}{4}} + e^{-j\frac{\pi k}{4}}\right) + \frac{d_1}{2}\left(e^{j\frac{\pi k}{2}} + e^{-j\frac{\pi k}{2}}\right)$$

$$= \frac{1}{8}\left(d_1 \cos\left(\frac{\pi k}{4}\right) + d_2 \cos\left(\frac{\pi k}{2}\right)\right)$$

BRUNNEN

Signale & Systeme

Multicarrier Transform

a) State if the proposal of the engineering team is correct.

$$\hat{x}(k) = x(k) \cdot \frac{1}{8},\ k = 0, \ldots, 7$$

Hence one period of the desired transient signal can be produced by an inverse C-DFT, as proposed by the team: correct → $x(k) = C\hat{x}(k)$

c) How many additional real

$x(0) = d_4$

$x(1) = d_1/2$

$x(2) = d_2/2$

$x(3) = d_3/2$

$x(4) = d_5$

$x(5) = d_3/2$

$x(6) = d_2/2$

$x(7) = d_1/2$

$$\rightsquigarrow \hat{x}(k) = d_1 \cos\left(\frac{\pi k}{4}\right) + d_2 \cos\left(\frac{\pi k}{2}\right) + d_3 \cos\left(\frac{3\pi k}{4}\right) + d_4 + d_5 (-1)^k$$

BRUNNEN

Signale & Systeme

Fast Fourier Transform

N-point DFT: $X(n) = \sum_{k=0}^{N-1} x(k)\, e^{-j 2\pi n k / N}$

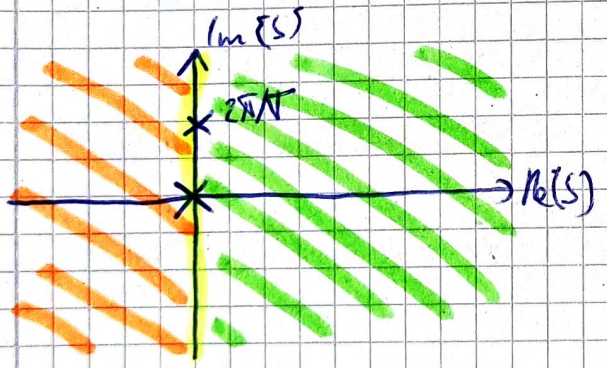
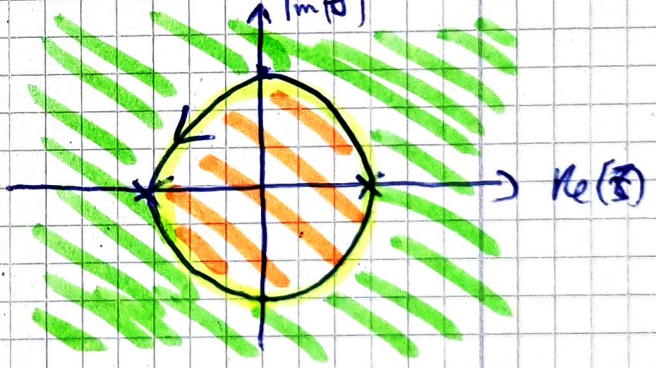
1) split sequence $x(k)$ into two with length $N/2$

z-Transform

Laplace Transform of Discrete Time Signals: The information about the time domain signal $x_c(t)$ is periodically repeated.

$z = e^{sT}$

↳ z-Transform: $X(z) = \mathcal{Z}\{x(k)\} = \sum_{k=-\infty}^{\infty} x(k)\, z^{-k}$

Laplace Transform	z-Transform
$X_c(s) = \sum_{x=-\infty}^{\infty} x(k)\, e^{-sTk}$	$X(z) = \sum_{k=-\infty}^{\infty} x(k)\, z^{-k}$
	
$\alpha < 0$	$\Rightarrow \lvert z \rvert < 1$
$\alpha > 0$	$\Rightarrow \lvert z \rvert > 1$
$\alpha = 0$	$\Rightarrow \lvert z \rvert = 1$

$\Omega = 2\pi f T_S \rightarrow f = \frac{\Omega}{2\pi T_S}$ Ω → Winkel in [rad]

BRUNNEN FH

Signale & Systeme

Laplace-Transf., DTFT, z-Transf.

Laplace:

$$X_\alpha(s) = \sum_{k=-\infty}^{\infty} x(k) e^{-s\nu k}$$

$\alpha = 0$ ⇓

DTFT:

$$X(\Omega) = \sum_{k=-\infty}^{\infty} x(k) e^{-j\Omega k}$$

$\Rightarrow$ $z = e^{s\nu}$

z-Transf.:

$$X(z) = \sum_{k=-\infty}^{\infty} x(k) z^{-k}$$

$z = e^{j\omega\nu} = e^{j\Omega}$

On the unit circle, z-Transf. & DTFT are identical!

The Laplace Transf. converges in a bad region parallel to the imaginary axis.
↳ The z-Transf. converges in a circular bad region!

BIBO

A system is stable, if $H(z)$ converges for $|z| = 1$

⇓

The convergence region of the z-Transform $H(z)$ of a stable system must include the unit circle

BRUNNEN

Signale & Systeme

Example

$$X(z) = \sum_{k=-\infty}^{\infty} x(k) z^{-k}, \text{ converges if } \sum_{k=-\infty}^{\infty} |x(k) z^{-k}| = \sum_{a=-\infty}^{\infty} |x(k) r^{-k}| < \infty$$

examples

$x(k) = \delta(k)$; $X(z) = \sum_{k=-\infty}^{0} x(k) z^{-k} = 1$

$x(k) = \varepsilon(k)$; $X(z) = \sum_{k=-\infty}^{0} \varepsilon(k) z^{-k} = \sum_{k=0}^{\infty} (z^{-1})^k = \frac{z}{z-1}$

$|z| > 1$

Geometric Series:

$$a_0 \sum_{k=0}^{N} q^k = \begin{cases} a_0 \dfrac{1-q^{N+1}}{1-q} & ; q \neq 1 \\ a_0 (N+1) & ; q = 1 \end{cases}$$

$$a_0 \sum_{k=0}^{\infty} q^k = \frac{a_0}{1-q}, \; |q| < 1$$

Examples:

$x_a(k) = \begin{cases} a^k, & k \geq 0 \\ 0, & k < 0 \end{cases}$; $X(z) = \sum_{k=-\infty}^{0} x_a(k) z^{-k} = \sum_{k=0}^{\infty} a^k z^{-k} = \frac{z}{z-a}$

$|a| \geq |z| > |a|$

⇒ The representation of a signal by its z-Transform is only unique, if also the convergence region is given!

BRUNNEN

Signale & Systeme

Causal Signals

$x(k) = 0$ for $k < k_0, k_0 \geq 0 \quad \overset{\mathcal{Z}}{\circ\!\!-\!\!\bullet} \quad X(z) = \sum_{k=k_0}^{\infty} x(k) z^{-k}$

Converges for $|z| \geq |z_0|$

The z-Transform of causal signals converges in an area outside a circle!

Some Causal signals:

$x(k) = 0$, for $k < 0$

$(\pm a)^k$	$\overset{\mathcal{Z}}{\circ\!\!-\!\!\bullet}$	$\frac{z}{z \pm a}$
$k a^k$	$\circ\!\!-\!\!\bullet$	$\frac{z \cdot a}{(z-a)^2}$
$k^2 a^k$	$\circ\!\!-\!\!\bullet$	$\frac{za(z+a)}{(z-a)^3}$
$\sin(\alpha k)$	$\circ\!\!-\!\!\bullet$	$\frac{z \sin(\alpha)}{z^2 - 2z\cos(\alpha) + 1}$
$\cos(\alpha k)$	$\circ\!\!-\!\!\bullet$	$\frac{z(z - \cos(\alpha))}{z^2 - 2z\cos(\alpha) + 1}$

BRUNNEN

Signale & Systeme

Properties of Z-Transform

$x(k-k_0) \circ\!-\!\bullet\; \mathcal{Z}\{x(k-k_0)\} = z^{-k_0} X(z)$

$x(k) \longrightarrow \boxed{E^{-1}} \longrightarrow x(k-1)$

$x(k) * y(k) \circ\!-\!\bullet\; X(z)\,Y(z)$

$x(k)\,y(k) \circ\!-\!\bullet\; \frac{1}{2\pi}\int_0^{2\pi} X(re^{j(\beta-\alpha)})\,Y(e^{j\alpha})\,d\alpha$

$\mathcal{Z}\{\alpha x(k) + \beta y(k)\} = \alpha X(z) + \beta Y(z)$

$\mathcal{Z}\{e^{\alpha k T} x(k)\} = X(e^{\alpha T} z)$

$\mathcal{Z}\{k \cdot x(k)\} = -z \frac{d}{dz} X(z)$

BRUNNEN

Signale & Systeme

z-Transform (1)

a) Calculate $X(z)$ of

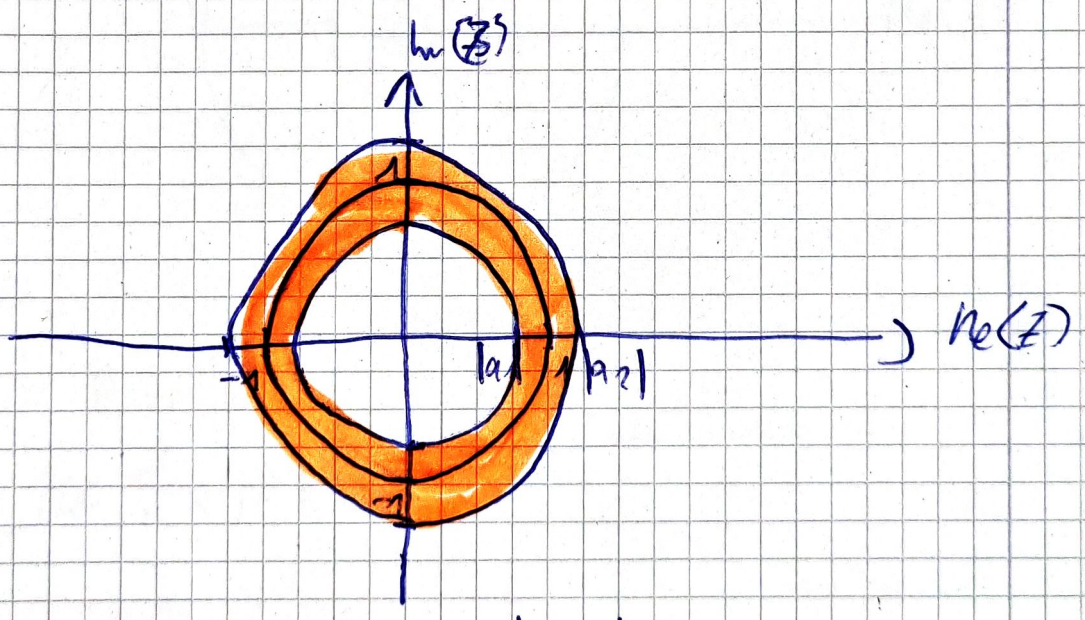
$$x(k) = \begin{cases} a_1^k & , k \geq 0 \,,\ a_1 \in \mathbb{R} \\ -a_2^k & , k < 0 \,,\ a_2 \in \mathbb{R} \end{cases}$$

$$X(z) = \sum_{k=-\infty}^{\infty} x(k) z^{k} = \sum_{k=-\infty}^{-1} -a_2^k z^{-k} + \sum_{k=0}^{\infty} a_1^k z^{-k}$$

$$= 1 - \sum_{k=0}^{\infty} a_2^{-k} z^{k} + \sum_{k=0}^{\infty} a_1^k z^{-k}$$

$$= 1 - \frac{1}{1 - \frac{z}{a_2}} + \frac{1}{1 - \frac{a_1}{z}} = 1 - \frac{a_2}{a_2 - z} + \frac{z}{z - a_1}$$

↳ $|z| < |a_2|$ ↳ $|z| > |a_1|$



$$|a_2| > |z| > |a_1|$$

$$\sum_{k=-\infty}^{-1} -a_2^k z^{-k} = 1 - \sum_{k=0}^{\infty} a_2^{-k} z^{k}$$

BRUNNEN

Signale & Systeme

z-Transform (1)

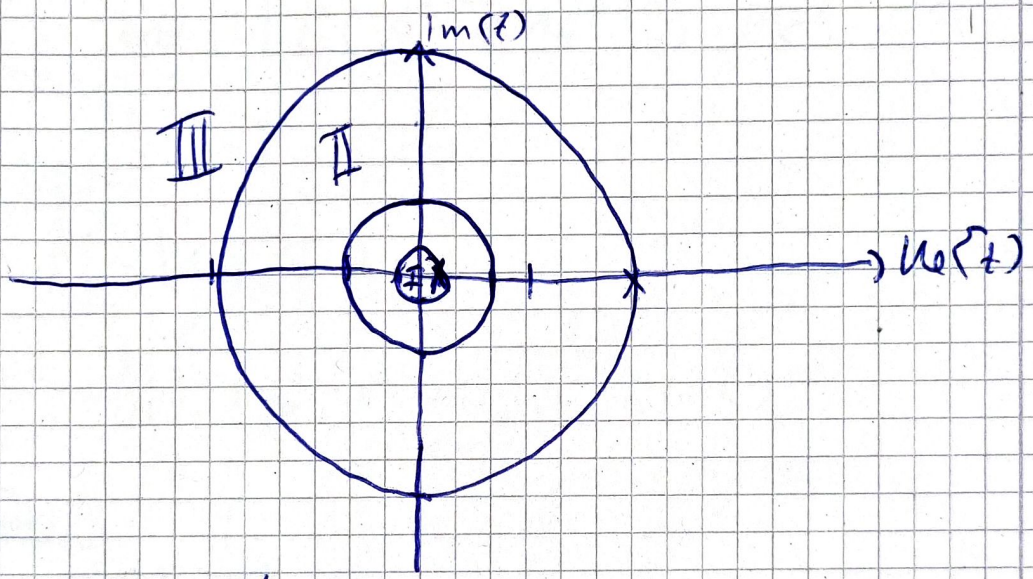
1) Causal Exponential sequence:

$$x(k) = a^k \varepsilon(k) \quad \circ\!\!-\!\!\bullet \quad X(z) = \frac{z}{z-a}, \quad |z| > |a|$$

2) Anticausal exponential sequence:

$$x(k) = -a^k \varepsilon(-k-1) \quad \circ\!\!-\!\!\bullet \quad X(z) = \frac{z}{z-a}, \quad |z| < |a|$$

3) $$H(z) = \frac{11z}{(z-1/4)(z-3)} = \frac{-1}{z-1/4} + \frac{11}{z-3}$$



I: $|z| < 1/4$

II: $\frac{1}{4} < |z| < 3$

III: $|z| > 3$

Poles Determine Area !

BRUNNEN

Signale & Systeme

z-Transform (1)

c)

1) $z^{-1}\frac{z}{z-a} = \frac{1}{z-a} \circ\!\!-\!\!\bullet\; a^{k-1}\varepsilon(k-1),\ |z|>|a|$

2) $z^{-1}\frac{z}{z-a} = \frac{1}{z-a} \circ\!\!-\!\!\bullet\; -a^{k-1}\varepsilon(-k),\ |z|<|a|$

Teil I:

$$H(z) = \frac{-1}{z-1/4} + \frac{12}{z-3}, \quad |z|<1/4$$

z | $\circ$

$$h(t) = -\left(\frac{1}{4}\right)^{k-1}\varepsilon(-k) - 12\cdot(3)^{k-1}\varepsilon(-k)$$

Teil II:

$$\frac{1}{4} < |z| < 3$$

$$h(t) = -\left(\frac{1}{4}\right)^{k-1}\varepsilon(k) - 12\,(3)^{k-1}\varepsilon(-k)$$

Teil III:

$$|z| > 3$$

$$h(t) = -\left(\frac{1}{4}\right)^{k-1}\varepsilon(k-1) + 12\,(3)^{k-1}\varepsilon(k-1)$$

BRUNNEN

Signale & Systeme

z-Transform (2)

a) Verify $Z\{x(k-k_0)\} = z^{-k_0} X(z)$

$$Z\{x(k-k_0)\} = \sum_{k=-\infty}^{\infty} x(k-k_0)\, z^{-k}$$

$k' = k - k_0$ (→)

$$= \sum_{k=-\infty}^{\infty} x(k')\, z^{-(k'+k_0)}$$

$$= z^{-k_0} \sum_{k=-\infty}^{\infty} x(k') z^{-k'} = z^{-k_0} X(z)$$

BRUNNEN